Sample preparation protocol for total proteomic analysis of mouse tissues including brain

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1) Materials

1.1) Consumables:

1.1.1) 1.5 ml protein low bind eppendorf tubes (Eppendorf™ #022431081)

1.1.2) Marker pen

1.1.3) Pipette set (1 ml, 200 µl, 100 µl, 20 µl, 10 µl)

1.1.4) Pipette tips low binding (1 ml, 250 µl, 10 µl, Star labs Bevelled tips refill # S1111-3700, S1111-1706, S1111-6700)

1.1.5) PPE kit (Lab coat, gloves, safety glasses)

1.1.6) Dry ice

1.1.7) Liquid Nitrogen

1.1.8) Ice bucket

1.1.9) 1.5 ml eppendorf tubes rack

1.1.10) 96 well plate- clear (Geneier Bio-one #655101)

1.1.11) 2 ml tubes (Axygen™ MCT200O)

1.1.12) 16-gauge needle (# Z261378. Sigma Aldrich)

1.1.13) Spray duster (Qconnect #KFO4499)

1.1.14) X100 20 mL Amber Glass EPA Vial (Cole Parmer # 11533750)

1.1.15) X72 40 mL Amber class EPA vial W Cap and seal (Cole Parmer # 10572553) (Note: Prepare all stock and working reagents in these amber vials to store as per the protocol)

1.1.16) Millipore pH Strips (VWR # 1.09584.0001)

1.1.17) PTFE- O rings (To place the stage-tip into the Eppendorf tubes. Generally, you could get from NEST group desalting columns and re use them https://www.nestgrp.com/)

1.1.18) S-Trap mini columns ((https://www.protifi.com/))

1.1.19) Sep-Pak Vac 1cc (50 mg) tC18-Cartridges (Waters # WAT054960)

1.1.20) XBridge BEH C18 Column, 130A, 3.5 um, 4.6 x 250 mm (Waters # 186003943)

1.1.21) 96 well 2 ml deep well plates (Fisher Scientific # 10680763)

1.1.22) Acclaim pepmap 100 100um*cm nano viper trap column (Thermo Fisher Scientific # 11312263)

1.1.23) Easy-Spray PepMap RSLC C18 2um, 50cmx75um (Thermo Fisher Scientific #ES802)

1.2) Reagents:

1.2.1) SDS Lysis Buffer: Final 2% (by mass) SDS in 100 mM Triethylammonium bicarbonate pH 8.5 (TEABC, this is the natural pH of this buffer and made from a 1 M TEABC stock purchased from Sigma Catalogue number# T7408-500 mL), 1 mM sodium orthovanadate, 50 mM NaF, 10 mM β- glycerophosphate, 5 mM sodium pyrophosphate, 1 μg/ml microcystin-LR, and complete EDTA-free protease inhibitor cocktail (Roche)

1.2.2) BCA protein assay kit (Pierce # 23225)

1.2.3) Tris (2-carboxyethyl) phosphine (TCEP) (Sigma Aldrich # 75259-10G). (Note: Prepare and store 10 μl aliquots of 1 M TCEP in Milli-Q H2O. Prior to use dilute the 1M TCEP solution 10 x in 300 mM TEABC to generate a stock solution of 0.1 M TCEP in 300 mM TEABC).

1.2.4) Ortho-Phosphoric acid 85% (by vol) (Sigma #5438280100). (Note: Prepare 12% (by vol) stock aqueous phosphoric acid by diluting in water and store in 4ºC)

1.2.5) S-Trap protein binding buffer (90% (by vol) aqueous LC grade methanol containing a final concentration of 100 mM TEAB, pH 7.1, made from a 1 M TEABC stock purchased from Sigma Aldrich # T7408-500 mL)

1.2.6) Sequencing grade trypsin (5 X 20 ug pack. Promega #V5111). (Note: Store stocks in -20ºC freezer and thaw trypsin stock just before the digestion step)

1.2.7) Methanol (VWR # 1.06035.2500)

1.2.8) LC-MS grade Acetonitrile (VWR # 1.00030.2500)

1.2.9) Iodoacetamide (Sigma # I1149)

1.2.10) LC grade Formic acid (Sigma # 695076)

1.2.11) Trifluroacetic acid (Sigma # 302031) (Note: Prepare and store 20% (by vol) aqueous TFA stock at 4ºC)

1.2.12) Empore C18 disks, 47 mm (CDS analytical #2215) (Note: Prepare a single layer with 16-gauge needle and pass it with spray duster into the 250 μl tip for 0.1 to 5 ug of peptide amount. For more than 5 ug use 2 or 3 layers of C18 material. Refer Figure 1 -see below for Stage-tip assembly Protocol Io staff could you insert figure 1 here?-Thanks).

1.2.13) 11 plex TMT-Kit (Thermo Fisher Scientific# 90110 and A37725) or 16 plex TMT-Kit (Thermo Fisher Scientific #A44520) depending on number of samples being analysed)

1.2.14) Anhydrous Acetonitrile (Sigma Aldrich #271004)

1.2.15) 50% (by vol) Hydroxylamine by mass (Sigma Aldrich # 467804)

1.2.16) 20% (by mass) aqueous SDS stock

1.2.17) LC buffer (0.1% (by vol) Formic acid in 3% (by vol) Acetonitrile)

1.2.18) Solvent-A1 (0.1% (by vol) TFA)

1.2.19) Solvent-A2 (0.1% (by vol) Formic acid

1.2.20) Solvent-B1 (50% (by vol) acetonitrile 0.1% (by vol) TFA)

1.2.21) Solvent-B2 (60% (by vol) acetonitrile 0.1% (by vol) Formic acid)

1.3) Equipment:

1.3.1) Pulveriser kit (https://cellcrusher.com/)

1.3.2) -80°C deep freezer, -20°C freezer and 4°C fridge

1.3.3) Benchtop centrifuge (VWR)

1.3.4) Milli-Q water system

1.3.5) Orbital shaker

1.3.6) pH meter

1.3.7) Plate reader for Protein quantification (BioTek Epoch)

1.3.8) Diagenode Bioruptor plus sonication system

1.3.9) Eppendorf Thermomixer

1.3.10) Thermo Savant Speedvac system (#SPD140DDA)

1.3.11) 1.5 ml tube floaters

1.3.12) Branson water bath sonicator

1.3.13) Dionex RSLC 3000 nano-LC system

1.3.14) Dionex RSLC 3000 LC system for Offline fractionation with Auto sampler or Fraction collector, micro pump and VWD detector

1.3.15) Orbitrap Fusion Lumos Tribrid Mass spectrometer

1.3.16) Thermo Savant Speed vac system (#SPD140DDA)

1.3.17) Nanodrop 1000 (Thermo Fisher Scientific)

1.3.18) Rubber bulb # Fisher brand™ Rubber Pipette Bulb# 12446180.

2) Protocol:

2.1) Protocol for S-Trap assisted digestion

2.1.1) Snap freeze mouse tissue immediately after isolation in liquid nitrogen and store at -80 ^{o}C.

2.1.2) Pulverize frozen tissue in liquid nitrogen to a fine powder and snap freeze immediately in 2 ml tubes. This is done using cell crusher kit.

(Note: Maintain all Pulverizer units in liquid nitrogen including spatula. After use wash the pulveriser with a tap water and clean thoroughly several times with 70% (by vol) ethanol)

2.1.3) Weigh pulverized tissue powder on a weighing balance (5-7% of the weight should be protein i.e. ~ 5 to 7 mg protein/100 mg)

(Note: Keep samples on dry ice during after weighing. Take a 1.5 ml tube check the weight on weighing balance and subsequently tare the weight. Now take small scoop of sample using pre-chilled fine spatula to weigh the sample)

2.1.4) Immediately add 10 times of the weight of the tissue of 2% (by mass) SDS lysis buffer (Note For 100 mg of tissue powder please add 1000 µl of lysis buffer)

2.1.5) Put samples on an orbital shaker in cold room at 1000 rpm for 15 min

2.1.6) Boil samples at 95^{o}C for 10 min and allow to cool

2.1.7) Sonicate samples using a Diagenode Bioruptor (use it at high energy for 10 cycles (30sec-ON/30Sec-Off)

2.1.8) Clarify lysate by centrifuging at 20,800g for 30 min at 4^{o}C

2.1.9) Measure protein amount using the Bicinchoninic acid assay (BCA) method in triplicate at 1 to 10 and 1 to 20 dilutions-repeat analysis if readings are not close

2.1.10) Take 200 µg of protein for total proteomic analysis

2.1.11) Perform reduction by adding a 1 in 10 dilution of a solution of 0.1 M TCEP dissolved in 300 mM TEABC to bring final concentration of TCEP to 10 mM

2.1.12) Incubate on a Thermomixer for 30 min at 60^{o}C temperature with a gentle agitation

2.1.13) Bring tubes to **room temperature** and add a one in 10 dilution of freshly prepared 0.4 m iodoacetamide dissolved in water (Note it is critical that the samples are at room temperature prior to addition of iodoacetamide)

2.1.14) Incubate in dark on a Thermomixer at room temperature for about 30 min with a gentle agitation

2.1.15) Quench alkylation by addition of a 1 in 10 dilution of 0.1 M TCEP dissolved in 300 mM TEABC to bring final concentration of TCEP to 10 mM

2.1.16) Incubate on a Thermomixer for 20 min at room temperature with a gentle agitation

2.1.17) Add SDS to a final concentration of 5% (by mass) from 20% (by mass) SDS stock.

(Note: The lysate is already in 2% (by mass) SDS so supplement with a stock of 20% (by mass) SDS in order to bring the final SDS concentration to 5% (by mass).

2.1.18) Add a 1 in 10 dilution of 12% (by vol) phosphoric acid into the sample to make a final concentration of ~1.2% (by vol) phosphoric acid (Note for example, 5 µl to 50 µl)

2.1.19) Dilute the sample to in 7 times the current volume of the mixture in of S-Trap wash buffer (90% (by vol) methanol in 0.1 m TEAB pH 7.1 v/v) (for examples if sample volume is 50 µl add 300 µl of S-Trap wash buffer ((90% (by vol) methanol in 0.1 m TEAB pH 7.1 v/v)) (Note: If the lysate volume is now more than 50 µl, scale S-Trap wash buffer accordingly i.e. 7 times the volume of the lysate)

2.1.20) Prepare an S-Trap mini column in a 2 ml tube.

2.1.21) Add the diluted protein mixture to the column.

2.1.22) Centrifuge briefly to capture the protein particles. (1000g for 1 min)

2.1.23) Wash column with 400 µl of S-Trap buffer a total of 4 times (spin 1000g for 1 min between washes) (Note that the protein remains bound on the column and SDS and buffer components that effect trypsin digestion are removed)

2.1.24) Move the S-Trap column to a clean 2 ml tube for digestion

2.1.25) Add a 100 µl solution of freshly dissolved trypsin containing 13 µg of trypsin and 2 µg of Lys-C freshly dissolved in 100 mM TEAB (1:15): (Note we use 6.5µg of trypsin and 1 µg of Lys-C per 100 µg of protein)

2.1.26) Centrifuge briefly at 200g for 1 min

2.1.27) Collect flowthrough and reapply the trypsin solution back onto the column being careful to avoid air bubbles

2.1.28) Cap the tubes and incubate at 47ºC without shaking for 1.5 hour. (Note do not shake as this causes bubbles and damage the column)

2.1.29) Incubate a room temperature for another 2 h at room temperature

2.1.30) Add 80 µl of 50 mM TEAB then spin to elute and place the eluate in a new 1.5ml Eppendorf tube termed "eluate tube".

2.1.31) Next, add 80 µl of 0.15% (by vol) Formic Acid and spin to elute. Also add this eluate to the "eluate tube"

2.1.32) Finally, add 80 µl of 50% (by vol) Acetonitrile in 0.15% (by vol) formic acid and spin to elute. Also add this eluate to the "eluate tube" (Note 3 eluates should have been added to the eluate tube)

2.1.33) Take 1-2 µl of the combined eluate, vacuum dry and inject on MS to verify the digestion efficiency (Note: Analyse data with a 70 min gradient run on QE HF-X or Orbitrap Lumos mass spectrometer in a FT-FT-HCD mode. Search data with Proteome Discoverer 2.1 or 2.4 version. Determine the digestion efficiency by plotting number of missed cleavages. Zero missed cleavages should be >75% and single missed cleavages should be between 20-23%)

2.1.34) Vacuum dry the remaining peptide amount and store in -80ºC deep freezer until ready to undertake TMT labelling

2.2) Protocol for Tandem mass tags labelling:

2.2.1) Dissolve 800 µg of each of the TMT mass tag reagents within the 11 or 16-plex TMT reagent kit with 41 µl of 100% by vol anhydrous acetonitrile to obtain a 20 µg/µl concentration for each TMT reporter tag.

2.2.2) Leave them at room temperature for 10 min. Following, vortex and briefly spin at 2000g for 2 min (Note: Dissolved TMT reagents are prone to hydrolysis so immediately after aliquoting store remainder reagent in -80ºC deep freezer for long-term storage up to six months and try to avoid multiple freeze thaw cycles).

2.2.3) Dissolve lyophilized peptides in 50 µl of a mixture containing 38 µl of 50 mM TEAB buffer + 8 µl of 100% (by vol) anhydrous acetonitrile.

(Note: it is important to maintain a final 30% (by vol) of anhydrous Acetonitrile for an effective TMT reaction)

2.2.4) Keep samples on a floater and place it on a water bath sonicator for 10 min

2.2.5) Centrifuge samples at high speed 20,8000g at room temperature for 10 min

2.2.6) Transfer dissolved peptides into a 1.5 ml protein low-binding eppendorf tubes

2.2.7) Add 10 µl of 20 µg/µl TMT reagent i.e. 200 µg aiming for a 1:1 mass ratio of peptide:TMT reagent.

2.2.8) Give a gentle vortex and brief spin 2000g for 1 min.

2.2.9) Place samples on a Thermomixer and incubate at room temperature for 2 h with a gentle agitation, 800 rpm

2.2.10) Add another 50 µl of 50 mM TEAB buffer to make a final 100 µl reaction. Vortex, brief spin at 2000g for 1 min and incubate on a Thermomixer for 10 min.

(Note: It is a good practice to maintain the total volume to 100 µl final reaction as it helps in reducing pipetting error when aliquoting 5 µl of sample for label check efficiency).

2.2.11) In order to verify the TMT labelling efficiency of each TMT mass tag, take a 5 µl aliquot from each of the TMT samples and pool this in a single tube and vacuum dry immediately using a SpeedVac (Note: It is important to verify the labelling efficiency of each TMT mass tag is and it should label > 98%, by analysing on Mass spec. We recommend doing this employing a 145 min FT-FT-MS2 study. This will establish that each reporter tag is efficiently labelled and ensure that an equal level of each peptide is labelled with each of the TMT tags. Search MS raw data with Proteome Discoverer 2.2 or 2.4 by enabling TMT-reporter tag mass (+229.163 Da) on Lysine residue and Peptide N-terminus as dynamic modifications. Filter TMT labelled Peptide spectral matches (PSMs) in the modification tab to calculate the number of labelled and unlabelled PSMs to determine the labelling efficiency. Also, export PSM abundance in txt.file, to plot a Boxplot using R-software to determine the ~1:1 abundance within and between replicates)

2.2.12) Place remaining 95 µl of the reaction in -80 freezer. If the labelling efficiency is >98% and levels of each labelled peptide appear to be close to 1:1 then proceed with the below steps.

2.2.13) Thaw stored TMT labelled samples from step 2.2.12 to room temperature

2.2.14) Prepare 5% (by vol) final Hydroxyl amine solution by dissolving in water from a 50% (by vol) stock solution

2.2.15) Add 5 µl of 5% (by vol) Hydroxylamine to each sample to quench TMT reaction by incubating the reaction at room temperature on a Thermomixer for 20 min

2.2.16) Pool all samples into a single tube.

2.2.17) Take 20% of the reaction i.e. 220 µl to as a backup, snap freeze on dry ice and vacuum dry (Note this is important because if there is a sample loss during the downstream analysis or to further validate).

2.2.18) Snap freeze the remaining 880 µl reaction and vacuum dry using Speed vac, for high pH fractionation.

2.2.19) Desalt sample using tC18 50 mg cartridge by following the protocol described in section 2.5

2.2.20) Submit sample to MS facility for high pH fractionation and request to fractionate to 96 fractions and concatenate by pooling distant fractions e.g A1+D1, A2+D2.. B1+E1, B2+E2 and so on to a total of 48 fractions for LC-MS/MS analysis (Note: Deep fractionation significantly reduces ion interference and ratio distortion caused by co-elution of peptides in turn and improves TMT-reporter ion quantification thus increasing the reliability of peptide and protein quantifications. Also, deep fractionation greatly improves the dynamic range of peptides/proteins and increases the coverage. We generally identify between 10,000 to 12,000 quantified protein groups).

2.2.21) For each of the 48 fractions measure peptide concentrations using a Nanodrop 1000 spectrophotometer

2.2.22) Prepare 2 µg of each fraction in 15 µl in LC buffer (0.1% (by vol) formic acid in 3% (by vol) Acetonitrile) and submit each fraction to the mass spectrometry facility.

2.2.23) Analyse each fraction by acquiring data in FT-FT-FT (MS3) HCD mode on a Orbitrap Fusion Lumos Mass spectrometer for 85 min run for each fraction

2.3 C18 stage-tip protocol:

2.3.1) Prepare single layer of C18 stage-tip using 16-gauge syringe needle

2.3.2) Dissolve the vacuum dried peptides sample derived from section 2.2.11 in 80 µl of solvent-A1 (0.1% (by vol) TFA)

2.3.3) Add 80 µl of 100% (by vol) Acetonitrile to the C18 stage-tip (Note this is required to activate the C18 resin).

2.3.4) Centrifuge at 2000g for 2 min at room temperature. Discard flow through

2.3.5) Add 80 µl Solvent-A1 0.1% (by vol) TFA (by vol) (Note this is required to equilibrate the C18 resin).

2.3.6) Centrifuge at 2000g for 2 min at room temperature. Discard flow through and repeat this step

2.3.7) Load the acidified peptide digest to Stage-tip.

2.3.8) Centrifuge at 1500g for 5 min at room temperature.

2.3.9) Reapply the flow through to the C18 stage-tip column and repeat this step

2.3.10) Add 80 µl of solvent-A1 (0.1% (by vol) TFA v/v), and centrifuge at 2,000g for 2 min at room temperature. Discard flow through. Repeat again.

2.3.11) Elute peptides absorbed to C18 column by placing the stage-tips into **new** 1.5 ml low binding tubes. (Note using new tubes is important to avoid contamination)

2.3.12) Add 30 µl of Elution buffer (Solvent B1- 50% (by vol) acetonitrile in 0.1% (by vol) TFA).

2.3.13) Centrifuge at 1,500g for 2 min. Add another 30 µl of Elution buffer (Solvent B1- 50% (by vol) acetonitrile in 0.1% (by vol) TFA) and repeat the centrifugation.

2.3.14) Immediately snap freeze on dry ice the eluates and vacuum dry the samples using a Speed-Vac

2.3.15) Submit to Mass spectrometry analysis as described in step 2.2.11

2.3) Sep-Pak purification protocol:

2.4.1) Dissolve vacuum dried pooled TMT sample from section 2.2.18 in 500 µl of solvent-A1 (0.1% (by vol) TFA)

2.4.2) Place sample on a floater and place it in on a water bath sonicator and sonicate it for 10 min

2.4.3) Centrifuge sample at 20,800g for 10 min and transfer sample into a new 1.5 ml eppendorf tube.

2.4.4) Check that the pH of the sample is acid >2, by pipetting 1 µl of the sample onto a pH strip (Note this is important because Stage-tips are equilibrated to acidic pH and this aid in proper binding of acidified peptides to C18 resin, if sample appears to be not acidic increase TFA concentration to 0.15% (by vol) and recheck pH).

2.4.5) Prepare single tC18 50 mg Sep-Pak cartridge and place this column in 15 ml falcon tube

2.4.6) Add 1 ml of 100% (by vol) Acetonitrile. Centrifuge on an Eppendorf centrifuge at 60g for 1 min and discard flowthrough. Repeat this step again (Note this step is required to activate the C18 resin).

2.4.7) Equilibrate the tC18 50 mg Sep-Pak cartridge, by adding 1 ml of solvent-A1 (0.1% (by vol) TFA v/v). Centrifuge on an Eppendorf centrifuge at 60g for 1 min and discard flowthrough. Repeat this step two more times

2.4.8) Load acidified sample onto the tC18 50 mg Sep-Pak cartridge. (Note: Do not centrifuge and let the sample pass through by gravity or if needed push sample using rubber bulb # Fisher brand™ Rubber Pipette Bulb# 12446180.)

2.4.9) Collect the flow through and reapply to the column and save the flow through (Note : If you don't detect any peptides in MS, the flow through should have the peptides and this has been caused due to poor equilibration of the column)

2.4.10) Wash the tC18 50 mg Sep-Pak cartridge by addition of 1 ml of solvent-A2 (0.1% (by vol) Formic acid). Centrifuge on an Eppendorf centrifuge at 60g for 1 min and discard flowthrough. Repeat this step two more times

2.4.11) Place column on a new 1.5 ml of Eppendorf tube. (Note using new tubes is important to avoid contamination)

2.4.12) Elute the peptides from the tC18 50 mg Sep-Pak cartridge by addition of 300 µl of solvent-B2 (60% (by vol) Acetonitrile in 0.1% (by vol) Formic acid). (Note: Do not centrifuge and let the sample pass through by gravity or if needed push sample using rubber bulb (Fisher brand™ Rubber Pipette Bulb# 12446180).

2.4.13) Repeat elution step two more times.

2.4.14) Combine the 900 µl of eluate and snap freeze and vacuum dry using Speed Vac

2.4.15) Store sample in -80ºC freezer or submit to mass spectrometry facility for high pH fractionation as described in step 2.2.20

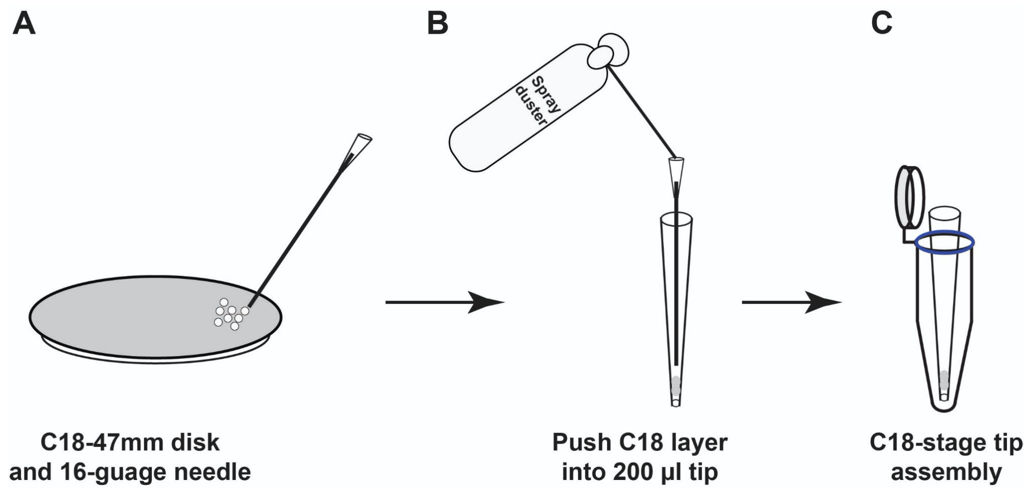

Figure1) Homemade C18 stage-tip assembly: A) Punch a C18 layer using 16 gauge needle B) Push the C18 layer into a 200 µl pipette tip using a spray duster C) Stage the tip using PTFE-O ring into a 1.5ml Eppendorf tube for desalting.